On the Separation of Harmfulness and Refusal in Reasoning Language Models

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Abstract

Recent work suggests that safety behavior in language models is not governed by a single latent signal: models can encode whether a request is harmful separately from whether they refuse it. We study whether this separation persists in reasoning or "thinking" models, where a chain of thought intervenes between the user request and the final answer. Across Qwen3.5 thinking, non-thinking, and stripped-template settings, we find that refusal behavior remains strong on harmful prompts while harmless over-refusal remains low. At the representation level, the harmfulness/refusal lens continues to organize model behavior, but refusal-related information is less cleanly localized at a single post-instruction boundary and instead interacts with think-template and reasoning positions. These results suggest that harmfulness/refusal separation remains useful for reasoning models, while simple single-token accounts of refusal become incomplete.

1. Introduction

Safety-aligned large language models (LLMs) are expected to answer benign requests while refusing harmful ones. Failures in either direction matter: under-refusal creates unsafe compliance, while over-refusal blocks useful benign requests (**??**). Prior work often assumes that harmfulness is encoded by the refusal direction (inster cite), but Zhao J. showed that LLMs encode harmfulness and refusal separately on instruct models (**?**).

Instruct models are trained to expect a chat template in order to recognize messages; Figure **??** illustrates the chat template for different configurations on Qwen3.5 (insert cite). In this setting, the authors define **tinst** as the last token of the user query, and **tpost** as the last token of the prompt tokens. After finding that stripping the tpost token induces 22% more accepted harmful queries, the authors conducted several experiments to show that the tinst token's activations encode harmfulness to a greater magnitude (replicated in Section **??**), whereas the tpost token encodes refusal to a greater magnitude. It was later causally shown that these two directions are distinct (replicated in Section **??**).

Reasoning models make this picture less straightforward. A thinking model may identify harmfulness early, reason about policy or intent during its CoT, and only later decide how to answer. This raises the question we study: **does harmfulness/refusal separation still hold in thinking models, and where does refusal live once reasoning tokens are introduced?**

For this matter, we search on thinking models on two different generation configurations: generation with a reasoning trace and generation without reasoning trace. Figure **??** rows a and b show the tokenizer template for these two configurations.

2. Datasets and Setup

In this section, we explain our experiments' datasets and configuration.

We first reproduced the original paper's experiments on Qwen2-7B (insert cite) and Llama3-8B (insert cite). After that, we ran our experiments on Qwen3.5-9B. We use the recommended sampling configuration, see Appendix (insert cite). For the judge model, we use Qwen3.5-4B, with few-shot examples containing thinking traces to reduce overthinking (insert cite).

In order to extract the harmful and refusal activations, we collect 3 datasets containing harmful queries: AdvBench (insert cite), SorryBench (insert cite) and JBB (insert cite), and 2 datasets containing harmless queries: XTest (insert cite) and Alpaca (insert cite). After that, we conduct the two-step pipeline shown in Figure **??**:

Dataset manual bucketing. The datasets are first combined according to their behavior (harmful or harmless) and generation configuration (thinking or non-thinking).

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		tinst (last query token) ↓	tpost (last prompt token) ↓
a)	Instruct model	<|im_start|> user \n q_1 q_2 ⋯ q_n	<|im_end|> \n <|im_start|> assistant
b)	Thinking generation	<|im_start|> user \n q_1 q_2 ⋯ q_n	<|im_end|> \n <|im_start|> assistant \n <think> \n
c)	Non-thinking generation	<|im_start|> user \n q_1 q_2 ⋯ q_n	<|im_end|> \n <|im_start|> assistant \n <think> \n \n </think> \n \n

Figure 1: Chat template tokenization on Qwen model family for instruct models (a) reasoning models thinking generation (b) and reasoning models non-thinking generation (b).

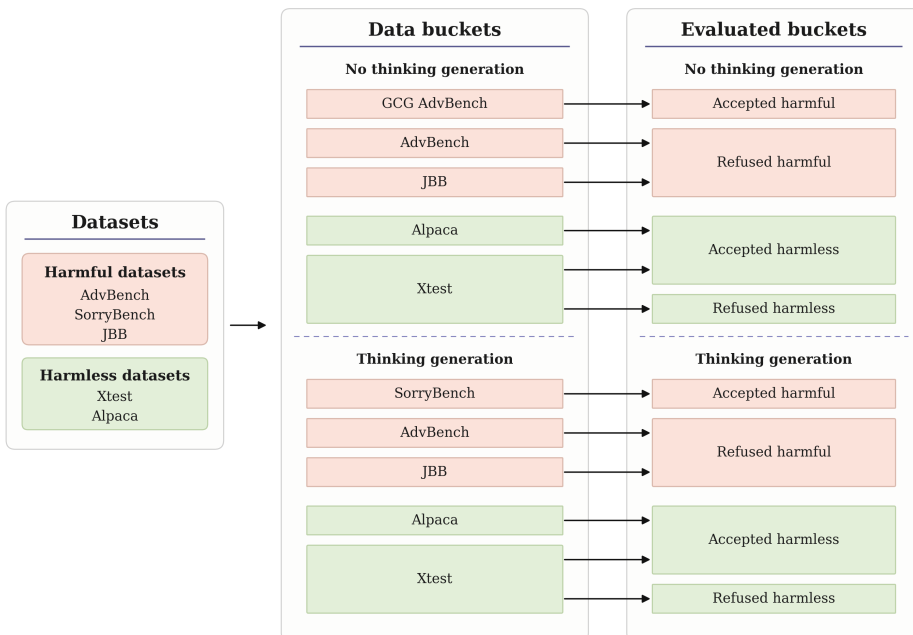


Figure 2: Two-step dataset configuration pipeline for the experiments. The datasets are first subject to a manual bucketing based on thinking or non-thinking generation, and then classified using a programmatic and later LLM-judged approach. Red buckets represent datasets containing harmful queries, whereas green buckets represent datasets containing harmless queries.

Response evaluation. After running the datasets' queries through the model, the generated responses are classified as refused (the model refused to answer due to safety concerns) or accepted (the model answered the query). The responses are first evaluated with a direct match against a common refusal sentence list (insert cite). After this, a judge model evaluates only the opposing behavior buckets, which are more likely to be misclassified (accepted harmful and refused harmless buckets).

We base our manual bucketing on the following basis: *(i)* Qwen3.5 has been trained with extensive safety alignment, so unlike the instruct model experiments, the model refuses almost all queries, so we compute a GCG (insert cite) jailbreak on the non-thinking generation with an ASR of 34% using nanoGCG (insert cite) to get genuine harmful responses; *(ii)* for the thinking generation, we use SorryBench, which contains soft-harmful queries that violate the model's safety guardrails but might not be as obscene as AdvBench and JBB; *(iii)* for refused harmless responses, we only use XTest because it contains harmful-looking harmless examples which might trick the model into a refusal.

3. Methodology

In this section, we present our approach with the experiments we conducted. We first test in Section **??** whether stripping the last tokens also decreases refusal like in Instruct models. Since that proves not effective, we then search for potential tokens with a strong harmful and refusal signal on Section **??**. We find that the tpost and gen1 tokens encode our desired behaviour on the non-thinking configuration. We confirm this belief in Section **??**, clustering the buckets in a harmful and refusal score space. In Section **??** then we steer both directions individualy on harmless data points to elicit high refusal rates. We then try an inversion generation experiment in Section (insert cite) to causaly prove the direction separation. We fail to prove the separation, hence our conclusions.

3.1. Token stripping generation

We compare the refusal rate of 4 generation configurations on the AdvBench and JBB datasets. This corresponds to the thinking, non-thinking, and two stripped versions. For stripped version a, we only remove the last `\n\n` from the non-thinking configuration, which is codified as only one token. For stripped version b, we strip everything in the thinking configuration until the <|im_start|> token. Results are shown in Table **??**. Contrary to the Instruct models, none of the configurations shows a decrease in the refusal rate.

Generation Configuration	Refusal Rate
Thinking	98.6%
Non-thinking	98.8%
Strip A	97.3%
Strip B	98.9%

Table 1: Refusal rate by generation configuration.

3.2. Activation visualization

We plot each token's mean activations to search for harmfulness and refusal encoding patterns. Taking activations for a chosen token, we implement Equation **??**. We first compute the centroid of the correctly behaved clusters, $\mu_{\text{harmful}}^{\text{refused}}$ and $\mu_{\text{harmless}}^{\text{accepted}}$, using the bucketing explained in Section **??**. After this, s_l is defined as the similarity of the mean activations for the chosen token at layer l. All the buckets are divided into an equal ratio train and test sets, to ensure that no activation is compared against a mean that cointains it's own value.

$$s_l(h_l) = \text{cos\_sim}\left(h_l, \mu_l^{\text{refused harmful}}\right) - \text{cos\_sim}\left(h_l, \mu_l^{\text{accepted harmless}}\right) \tag{1}$$

We use the activations of the opposing expected behavior to test the following hypothesis: if a token's representation is closer to the cluster matching its harmfulness label than to the cluster matching its expected behavior, then that token encodes harmfulness more strongly than refusal, and vice versa. For example, if accepted harmful query activations are closer to the refused harmful centroid, then this token encodes harmfulness to a greater magnitude than refusal; if instead they are closer to the accepted harmless centroid, then the refusal direction is more present. The same logic applies to refused harmless activations.

We run this experiment on the following tokens: *(i)* t_{inst} and t_{post} tokens; *(ii)* \n and thinking tag tokens; *(iii)* a set of 15 selected tokens in the CoT, for which we take the first 5 tokens of the CoT, the last 5 tokens of the CoT, and 5 selected middle tokens — for the middle tokens, we divide the CoT into 5 groups (excluding the first and last 5) and take the middle index of each group; *(iv)* the first 3 generated tokens.

Figure **??** shows the results for the distinguished tokens, All the other tokens' plots can be found in the Appendix (cite here). Figure **??** shows the desired opposing behavior on Qwen2-7B, tinst encodes harmfulness while tpost encodes refusal. Figure **??** and Figure **??** show the activations for Qwen3.5-9B on thinking and non-thinking generations. Tpost on the non-thinking configuration is the only token on the prompt activations that show the desired behavior. For this reason, we choose tpost and the first generated token on the non-thinking configuration to continue with our experiments. As Figure **??** shows, we hypothesize that tpost contains the harmfulness direction and the first generated token contains the refusal direction.

The thinking configuration tokens show less promising results. Appendix (insert cite) shows all the tokens on this configuration. We try the \n token after the opening think token as the harmful token and mid1 token for the refusal direction, but we get no interesting results. We conclude that either there is no clear separation, or the SorryBench dataset was not harmful enough to capture the harmful direction.

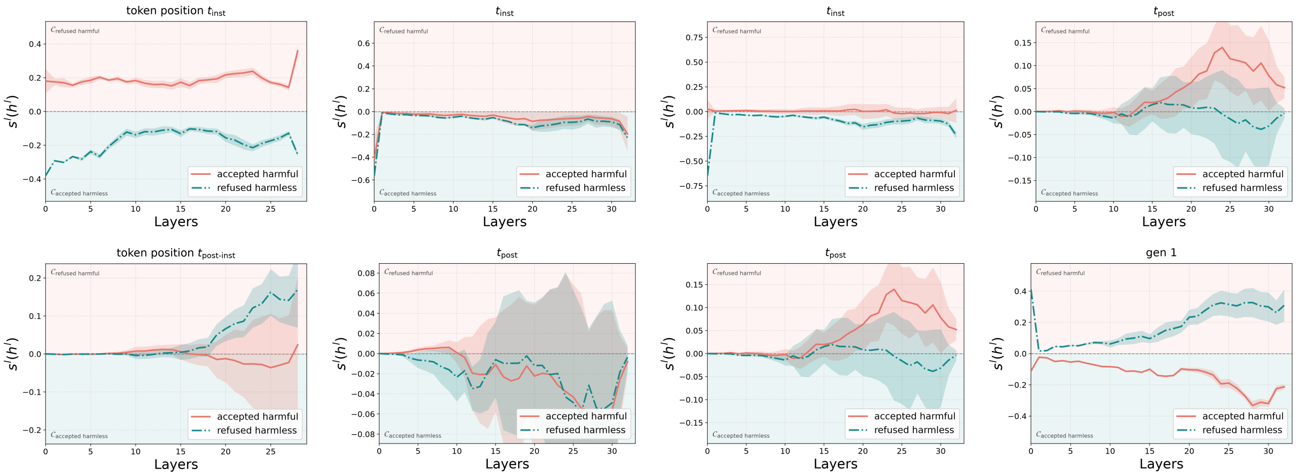
3.3. Correlation between beliefs of harmfulness and refusal replication

In this section, we plot the harmful score (Equation **??**) and refusal score **??** of each activation bucket for the non-thinking configuration using tpost as the harmful token and gen1 as the refusal token. We apply the same train and test split as in Section **??**

$$\Delta_{\text{harmful}} = \frac{1}{L}\sum_{l=1}^{L}\left(\text{cos\_sim}\left(h_l^{t_{\text{post}}}, \mu_{l,\text{ harmful}}^{t_{\text{post}}}\right) - \text{cos\_sim}\left(h_l^{t_{\text{post}}}, \mu_{l,\text{ harmless}}^{t_{\text{post}}}\right)\right) \tag{2}$$

$$\Delta_{\text{refuse}} = \frac{1}{L}\sum_{l=1}^{L}\left(\text{cos\_sim}\left(h_l^{t_{\text{gen1}}}, \mu_{l,\text{ refuse}}^{t_{\text{gen1}}}\right) - \text{cos\_sim}\left(h_l^{t_{\text{gen1}}}, \mu_{l,\text{ accept}}^{t_{\text{gen1}}}\right)\right) \tag{3}$$

Figure **??** shows the resulting clusters, the four clusters are clearly defined on it's corresponging space. We note that the magnitude of the score is nearly half as the replication for the Qwen2 model, suggesting that although both directions are present, the directions might not be as distinct as expected.



(a) Desired behavior, replicated on Qwen2-7B for tinst and tpost.

(b) Qwen3.5-9B on thinking configuration, tinst and tpost activations

(c) Qwen3.5-9B on non-thinking configuration, tinst and tpost activations

(d) Qwen3.5-9B on non-thinking configuration, tpost and gen1 activations

Figure 3

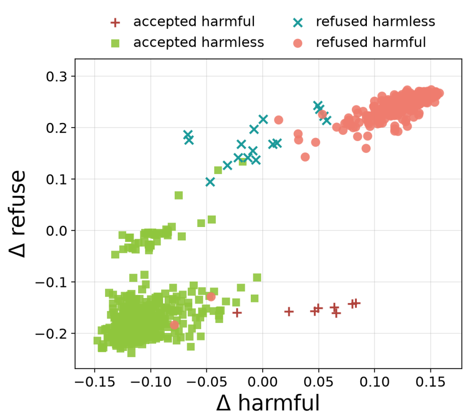


Figure 4: Non-thinking generation activations for tpost and gen1 tokens clustered according to refusal and harmful score.

3.4. Refusal elicitation

We test our tinst and gen1 tokens activations to check whether we can elicit refusal. For each layer, we extract the harmful vector substracting the tpost token's mean harmless activation from the tpost mean harmful activation (Equation **??**). We apply the same logic with the refusal vector in Equation **??**. Afterwards, the computed vector on each layer is added to the model's residual stream while generating a response, multiplied by a magnitude coefficient.

$$v_l^{\text{harmful}} = \mu_{l,\,\text{harmful}}^{t_{\text{post}}} - \mu_{l,\,\text{harmless}}^{t_{\text{post}}} \tag{4}$$

$$v_l^{\text{refuse}} = \mu_{l,\,\text{refuse}}^{t_{\text{gen1}}} - \mu_{l,\,\text{accept}}^{t_{\text{gen1}}} \tag{5}$$

$$h_l' = h_l + c * v_l^{\text{harmful}} \tag{6}$$

Figure **??** shows the steering results. We first conduct two baseline trials to measure the refusal rate induced by a random vector matched in coefficient and magnitude to the two vectors of interest. As the random vectors induce no refusal, we proceed to evaluate the harmful and refusal vectors. We observe a maximum refusal rate of 68% for the harmful direction at the 14th layer, and a maximum refusal rate of 36% for the refusal direction. Both directions successfully induce refusal in the responses, but to a lesser extent than in the Instruct model.

3.5. Harfmul and refusal direction causal separation

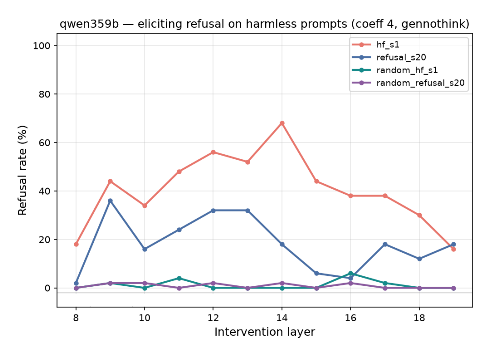


Figure 5: Refusal rate on harmless instructions after steering with the harmful (fh$_s$1)$andrefusal(refusal_s20)directions.random_hf_s1representsabase$

Behavioral buckets. We group examples by both prompt type and observed behavior: accepted harmful, refused harmful, accepted harmless, and refused harmless. These buckets let us distinguish whether a representation tracks input harmfulness, output refusal, or a mixture of both.

Harmfulness and refusal scores. Following the original paper, we compute cosine-difference scores between bucket means. The harmfulness score contrasts harmful and harmless anchors, while the refusal score contrasts refused and accepted anchors. For non-reasoning models these scores are naturally measured at t_{inst} and $t_{\text{post-inst}}$. For thinking models, we extend this to a 25-slot layout covering the instruction boundary, think-template tokens, sampled CoT positions, the </think> boundary, and early answer tokens.

Figures 2 and 3. Figure 2-style plots test where safety-relevant information is localized across layers and token positions. Figure 3-style plots test whether harmfulness and refusal behave as separate axes: if they are separable, accepted harmful examples should be high on harmfulness but low on refusal, while refused harmless examples should be low on harmfulness but high on refusal.

4. Conclusions

5. Limitations

Our work has the following limitations.

Tested models. The thinking experiments were executed on a single family model, Qwen3.5. Although both Qwen3.5-4B and Qwen3.5-9B were used, the resulted conclusions are only valid for Qwen family models.

Data scarcity. As stated on Section **??**, Qwen3.5 models are trained to refuse harmful instructions, so the GCG attack was necessary to generate accepted answers. Due to time constraints, our accepted harmful bucket only contained 17 data points on the non-thinking generation.

6. Related Work

Activation steering and refusal directions. Work on activation steering shows that behaviorally meaningful directions can be extracted from hidden states and used to modify model behavior without retraining (**??**). In safety settings, **?** show that refusal in chat models can often be mediated by a single residual-stream direction. This provides a natural starting point for studying refusal as a geometric object in activation space.

Harmfulness, refusal, and over-refusal. Our work directly builds on **?**, who argue that harmfulness and refusal are encoded separately rather than as a single safety axis. Related work complicates the geometry of refusal: refusal may occupy a concept cone or multiple directions rather than one vector (**??**), and over-refusal can be task-dependent even when harmful refusal is comparatively low-dimensional (**??**).

Reasoning models and CoT safety. Recent work suggests that reasoning changes the refusal mechanism. **?** show that where and whether reasoning models refuse depends on CoT content, and **?** find that simple refusal steering becomes less reliable once CoT is fixed. Other work shows that reasoning can create new jailbreak surfaces or alter safety decisions during generation (**???**). These findings motivate analyzing not only prompt-boundary tokens, but also thinking and answer-boundary positions.

Monitoring reasoning. Finally, probe-based monitoring work asks whether unsafe final behavior can be predicted before a model finishes thinking (**?**). Steering-vector analyses of reasoning behavior more broadly also suggest that latent directions can reveal structure inside thinking models (**?**).

A. Auxiliary Analyses

A.1. Jailbreak Geometry

As an auxiliary analysis, we project several jailbreak prompt families into the same Δ harmful / Δ refuse space used for the main representation analysis. The goal is not to replace a full judge-based safety evaluation, but to check whether qualitatively different jailbreak mechanisms occupy different regions of the harmfulness/refusal plane. The resulting geometry is informative because the attack families do not collapse into one generic "jailbreak" cluster.

The refused-harmful baseline occupies the upper-right region: both Δ harmful and Δ refuse are positive, matching the intended interpretation that the model recognizes the requests as harmful and refuses them. Template jailbreaks move in the opposite direction: they stay near positive Δ harmful but shift to negative Δ refuse, which is exactly the pattern expected from a jailbreak that does not erase the model's harmfulness representation but suppresses the refusal behavior. This mirrors the original paper's claim that jailbreaks can decouple harmfulness recognition from refusal.

The evil-persona prompts form a very different cluster. They have negative Δ harmful but positive Δ refuse, suggesting that the persona framing changes the prompt representation away from the ordinary harmful-instruction region while still eliciting a refusal-like state. In the heuristic audit, this family produces 50/50 refusals, so in this run the evil-persona wrapper appears to strengthen or preserve refusal rather than inducing compliance.

Table 2: Heuristic harmful-compliance audit for the jailbreak prompt families in **??**. The evaluator separates explicit refusals, benign diversions, and responses that appear to continue addressing the original harmful request. These labels are a sanity-check audit rather than a substitute for human or LLM judging.

Prompt family	n	Refusal	Benign	Harmful
Evil persona	50	50	0	0
GCG suffix	20	15	4	1
GPTFuzzer/template	50	0	23	27
Persuasion	50	8	31	11

The GCG suffix prompts are more heterogeneous. Some points lie near the template-jailbreak region with low or negative refusal, while others retain positive refusal. This spread suggests that suffix optimization does not create one uniform latent effect; instead, individual suffixes can push the model into different harmfulness/refusal regimes. The persuasion prompts mostly occupy the lower-left region, with negative Δ harmful and negative Δ refuse, consistent with many responses becoming non-refusal diversions rather than direct harmful compliance. Overall, the scatter supports using the two-axis harmfulness/refusal geometry to compare jailbreak mechanisms, not only clean harmful and harmless prompts.

A.2. CoT-Length and Projection Diagnostics

We also include an exploratory layer sweep from a separate Qwen3-thinking probe. For each intervened layer, we measured whether a chain of thought was produced, its length, and projection scores onto harmfulness/refusal directions at t_{inst} and $t_{\text{post-inst}}$. This diagnostic complements the main token-slot analysis by showing how inversion-style interventions co-vary with both CoT length and layer-wise projection scores. In the summarized runs, non-inversion settings produce longer CoTs on average, while inversion settings shorten the reasoning trace for both harmfulness and refusal probes. This gives a direct way to connect activation steering, reasoning length, and the internal harmfulness/refusal coordinates.

Citations and References

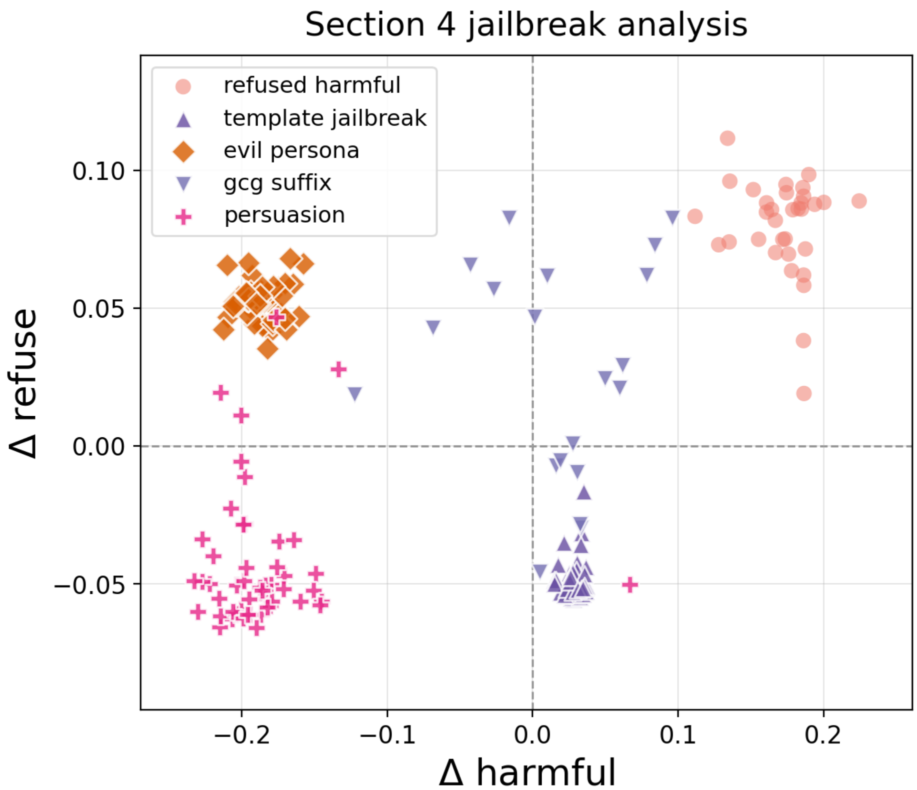


Figure 6: Exploratory jailbreak geometry for Qwen-7B. Refused-harmful baseline prompts are plotted alongside template jailbreaks, evil-persona prompts, GCG suffixes, and persuasion prompts. The attack families occupy distinct regions of the harmfulness/refusal plane.

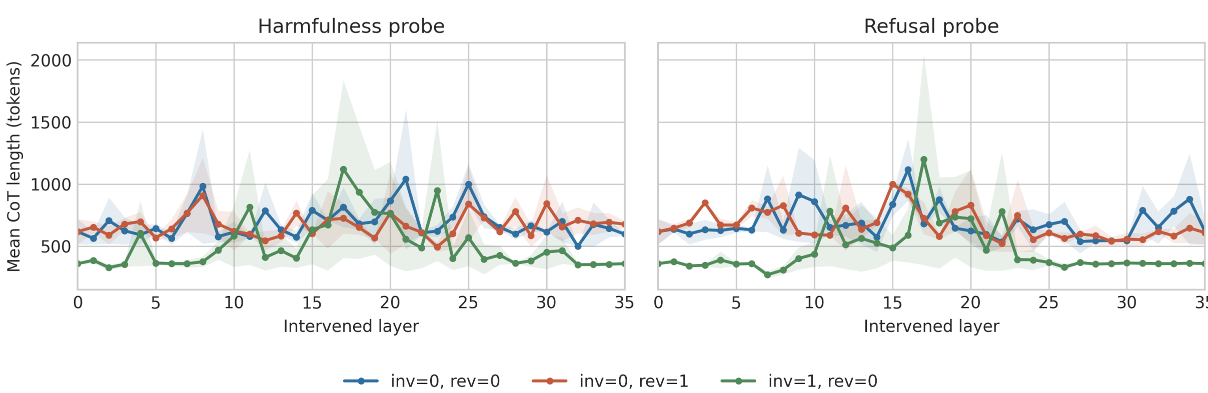


Figure 7: Exploratory CoT-length diagnostics under inversion and non-inversion settings. Inversion settings produce shorter CoTs on average than non-inversion settings for both harmfulness and refusal probes.

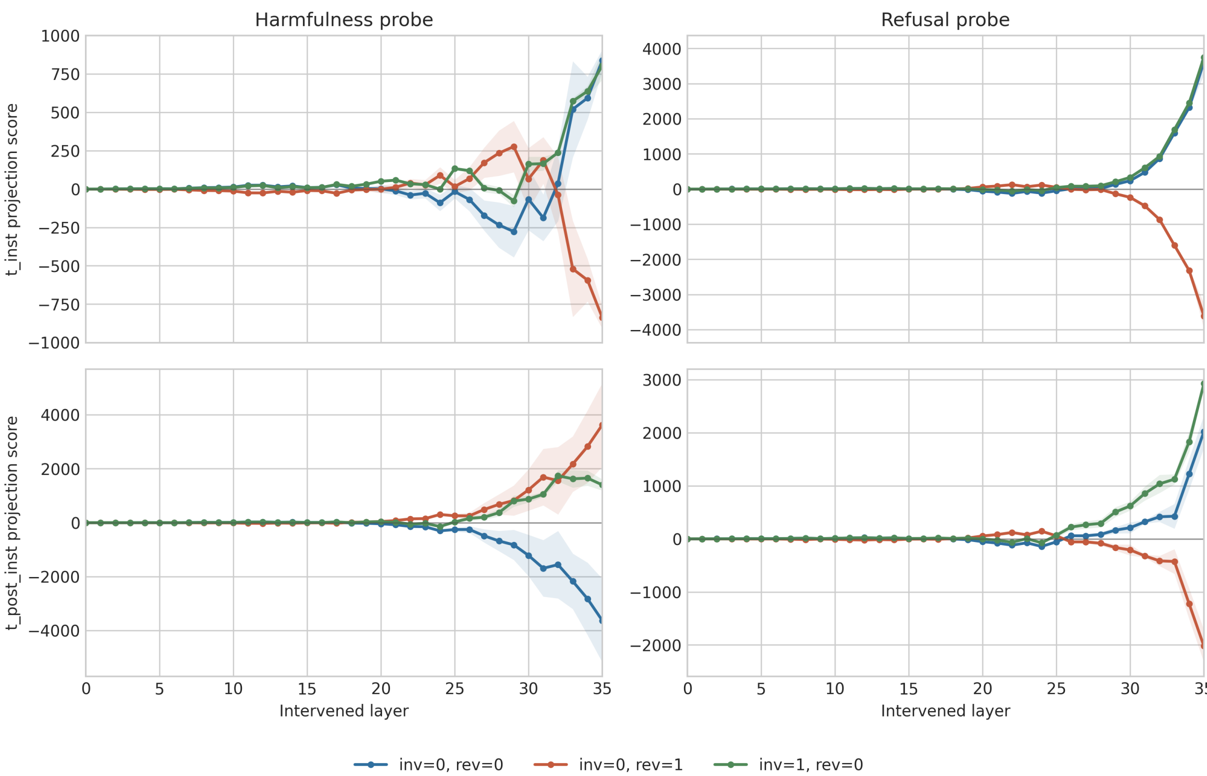


Figure 8: Exploratory projection-score layer sweep for harmfulness and refusal probes. Scores vary strongly by layer, especially in later layers, suggesting that steering effects are layer-dependent rather than uniform across the network.